



**SOFE4620 - Machine Learning and Data Mining
Winter 2025**

**Dr. Masoud Makrehchi
NBA Lineup Prediction Report
March 19, 2025**

Group Member

Name: Rishab Singh
Student ID: 100787473

Group Member

Name: Faisal Akbar
Student ID: 100846786

Group Member

Name: Saieashan Sathivel
Student ID: 100818528

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NBA Lineup Prediction for Optimized Team Performance

Introduction

The goal of this project is to create a machine learning model that forecasts the **optimal fifth player** for a home team in an NBA game based on partial lineup data and information and game-related attributes. Using the available historical NBA data from the years 2007 to 2015, the aim is to achieve maximum team performance through the analysis of lineup trends and forecasting the ideal player to add to a lineup.

The **notebook** organizes data exploration, data preparation, and model deployment. The main focus was on comprehending:

- Lineup stability and temporal change
- Number of occurrences of the player at different positions (home_0 through home_4).
- Lineup result analysis to identify winning lineups of players
- Typical matchups, where we establish who tends to play against specific away lineups.

These results guided the model selection and the feature engineering process to improve the predictive capabilities of lineup recommendations.

Data Collection and Preprocessing

Dataset Description

The project's dataset includes all the historical data of the NBA games from the years 2007 to 2015 recorded in CSV files. Also includes information regarding the teams' compositions, game outcomes and overall player statistics. These raw data were preprocessed using Pandas and NumPy libraries to clean and standardize them.

Feature Selection and Cleaning

This project involved certain dataset constraints, to comply with the constraints only the allowed features from each provided file were used such as the:

- Game metadata: game, season, home_team, away_team, starting_min.

- Player Lineups: home_0, home_1, home_2, home_3, home_4 and away_0, away_1, away_2, away_3, away_4.
- Outcome variable: outcome (if home team wins = 1, if away team wins = -1)

Steps for the data cleaning process:

Handling missing values: Entries with missing player or team information were removed.

1. **Standardizing team acronyms:** Team names were mapped to a unified format
For example: (NJN becomes BRK, NOH becomes NOP)
2. **Creating lineup tuples:** The sorted player IDs are found in tuples for modelling.
3. **Filtering for balance:** An equal number of positive (home win) and negative (home loss) outcomes were selected to use for the untrained training data.

Exploratory Data Analysis (EDA)

EDA was used to find the hidden patterns in the lineups to uncover hidden patterns in the lineups. Unique lineup counts per game were analyzed. The most frequently used home and away lineups were identified. Furthermore, the player position frequency was important and examined to detect any trends in the common lineup formations. Finally, outcome-based analysis gave insights on which NBA lineups would give a higher chance of winning games.

Model Selection and Training

Model Choice

This project involved the use of a supervised machine learning model, where the ultimate model to consider was:

- **Random Forest** – This machine learning algorithm is able to portray both linear and non-linear relationship between two variables

For example, we choose these models as they provide good support for handling categorical variables, establish intricate links between players, as well as are quite scalable for large datasets.

Training Process

Four of the following processes were included in the training process:

1. **Data Splitting:** The dataset was divided into **80% training data** and **20% testing data**.
2. **Feature Scaling:** Player statistics were standardized using StandardScaler to normalize numerical features.
3. **Cross-Validation:** K-fold cross-validation was applied to improve model robustness and prevent overfitting.
4. **Hyperparameter Tuning:** Grid search was performed to optimize model performance by adjusting tree depth, learning rate, and other parameters.

Problems Faced

One major issue our group faced was if we needed to remove a player from the home lineup during training to properly simulate game player substitutions in actual NBA games. This made sense to us but we faced potential overfitting of the model risks, the model can perhaps become too reliant on the other four remaining players therefore limiting generalization. To address this problem, we tested the impact of the player removed using different training configurations. Furthermore, the model's performance was evaluated on different lineups to see if it generalizes efficiently. Lastly, another approach was to include the training dataset with simulated player absences.

Evaluation and Results

For the evaluation and results of the project, several factors were assessed. The model's prediction was evaluated based on:

- **Accuracy Score:** The percentage of correct predictions for the fifth player.
- **Feature Importance:** Identifying which features contributed most to the predictions.
- **Historical Contribution:** Assessing how well the predicted player performed in past games with similar conditions.

Findings and Results

Overall, the model successfully recommended the optimal fifth player, aligning well with **historical team performance in similar games**. Certain lineup combinations consistently led to **higher success rates**, emphasizing the importance of **player synergy and tactical chemistry** in a team sport like basketball.

The accuracy results from model testing were as follows:

- **Top-1 Accuracy:** 40.6% – The model correctly predicted the most likely player 40.6% of the time.
- **Top-3 Accuracy:** 65.6% – The correct player appeared in the top 3 predictions in 65.6% of cases.
- **Top-5 Accuracy:** 75.4% – The correct player was among the top 5 choices in 75.4% of predictions.
- **NBA 2016 Accuracy:** 15%

These results highlight the **model's strong performance in lineup prediction**, with a significant improvement when considering multiple probable choices. The feature correlations revealed that a player's **past performance, position, and lineup stability** were among the strongest predictors for success.

Conclusion and Future Work

Project Success

The NBA lineup prediction model proved how machine learning technology can deliver significant value to sports analytics through accurate forecasting of the fifth starting member. The model delivered useful data regarding optimized team arrangements and player metrics through its decent identification of the best fifth team member.

1. Real-time game metrics such as player fatigue should be included together with recent form data alongside in-game statistics to improve prediction abilities.
2. The optimization strategies need expansion to study substitutions along with player rotations in a dynamic fashion.
3. The implementation of recurrent neural networks (RNNs) should be studied for their ability to analyze temporal player behaviour patterns.
4. Enhance dataset diversity: Incorporating more recent NBA data for better generalization.

The project demonstrates how decision-making that relies on data analysis enables sports management teams to achieve peak performance alongside enhanced strategic planning. Through machine learning methods coaches together with analysts and teams obtain valuable insights which help them design optimal strategic player deployments.